A Formal Lower-Bound for Numbers of the Form $p + 2^k + 2^l$ in Lean

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Abstract

We formalize and prove a short obstruction result for integers of the form $p + 2^k + 2^l$, where p is prime. The central statement shows that any integer represented by such a decomposition is bounded below by 4. This result is formalized in Lean 4 within the ARA legacy number-theory module and is used to derive elementary consequences for the corresponding exceptional odd set.

Our contribution is not a deep number-theoretic theorem, but rather a fully machine-checked mini-framework: a clear definition of representability, a lower-bound lemma proved by rewriting and arithmetic lemmas from arithmetic foundations, and two corollaries identifying explicit exceptional members in the odd set.

1 Introduction

Expressions of the shape $p + 2^k + 2^l$ with prime p and nonnegative integers k, l appear in additive problems that combine prime inputs with dyadic powers. A common first step in such investigations is to identify basic lower bounds and small exceptions before turning to structural conjectures or computational searches.

In this note we formalize a minimal shell for this pattern and prove a universal obstruction: every representable number is at least 4. As a direct consequence, small odd numbers fail to be representable, yielding a nonempty exceptional odd set. The contribution is intentionally foundational: it demonstrates a reusable Lean interface for this representation and establishes a formally verified base for further extensions.

The development is implemented in a modular style suitable for future generalization (e.g., to additional terms or constrained exponents), and is hosted in 'AraLibrary.NumberTheory.PrimeTwoPowers'.

2 Formalization and Definitions

We define a predicate on natural numbers

Definition 1. *$RepresentableByPrimeAndTwoPowers(n)$ asserts that there exist naturals p, k, l such that p is prime and $n = p + 2^k + 2^l$.*

The exceptional odd set is defined as $\{n \mid n \text{ is odd and } \neg RepresentableByPrimeAndTwoPowers(n)\}$. This set records odd naturals that do not admit the prescribed decomposition.

The project context is intentionally conservative: only number-theoretic core lemmas from Mathlib are used, and no additional axioms are introduced.

3 Main Result

The key theorem is a lower-bound lemma.

Theorem 1 (Lower bound for representable numbers)**.** *For any naturals* n, p, k, l*, if* p *is prime and* $n = p + 2^k + 2^l$*, then* $4 \leq n$*.*

The proof is short. From primality we obtain $2 \leq p$. For all $k, l \in \mathbb{N}$, $1 \leq 2^k$ and $1 \leq 2^l$. Summing these inequalities shows $4 \leq p + 2^k + 2^l = n$.

The theorem is mechanized with a direct rewriting step and linear arithmetic, minimizing dependence on higher-level lemmas.

4 Derived Corollaries

The lower bound yields immediate consequences.

Theorem 2. 1 *is not representable as* $p + 2^k + 2^l$*.*

Assuming a representation of 1, the lower-bound lemma gives $4 \leq 1$, a contradiction.

Theorem 3. 1 *belongs to the exceptional odd set.*

This follows by combining oddness of 1 with the previous theorem.

Theorem 4. *Every odd natural number* $n < 4$ *belongs to the exceptional odd set.*

For such n, oddness is by assumption. A hypothetical representation implies $4 \leq n$, contradicting $n < 4$.

The odd constraint is compatible with the definition of exceptional odd set, so the membership follows cleanly. A direct instantiation gives $1 \in \text{ExceptionalOddSet}$, and the same argument with odd $n = 3$ shows 3 is also exceptional as an immediate corollary.

5 Implementation Notes

The implementation uses three ingredients. First, explicit existential quantification for representations. Second, prime-specific arithmetic facts from Mathlib (‘Nat.Prime.two$_{l}e$‘).$Third, standardarithmeticlo$

The source is short, deterministic, and does not rely on specialized number-theory machinery, which makes it a good entry point for proof engineering and reusable formal components.

6 Discussion

Although elementary, this module supports a common pattern in formal number theory: defining a computationally meaningful class of numbers, proving robust guard lemmas by arithmetic, and extracting certified exceptional families. In this case, the bound $4 \leq n$ is a structural obstruction that immediately constrains the search space of candidate odd counterexamples.

As a research artifact, this result is most useful as a verified foundation for larger workflows. For example, one can add additional representation terms, parameter bounds, or congruence constraints and retain the same style of theorem shape. Future work may include automated generation of the exceptional-set witness lists for finite prefixes and formalized coverage results under stronger hypotheses.

7 Conclusion

We provide a formal, publication-ready mini-result in Lean: numbers of the form $p + 2^k + 2^l$ are all at least 4, and odd numbers below 4 are therefore in the exceptional odd set. The proof is concise, fully mechanical, and reproducible. This demonstrates the viability of building mathematically meaningful statements directly from a trusted formal core, and of using such statements as robust building blocks for larger conjecture-benchmark pipelines.

A Full Lean Formalization

The full development used in this manuscript is:

```
import Mathlib

/--!
Reusable shell for numbers representable as a prime plus two powers of two.

This module is extracted from the ARA Erd\H{o}s #9 campaign. It contains only the
small verified obstruction that any representation ‘p + 2^k + 2^l‘ is at least
‘4‘, plus immediate consequences for the exceptional odd set.
-/

namespace AraLibrary
namespace PrimeTwoPowers

/-- Numbers representable as ‘p + 2^k + 2^l‘ with ‘p‘ prime. -/
def RepresentableByPrimeAndTwoPowers (n : \N) : Prop :=
  \exists p k l : \N, Nat.Prime p \land n = p + 2 ^ k + 2 ^ l

/-- Odd numbers not representable as ‘p + 2^k + 2^l‘. -/
def ExceptionalOddSet : Set \N :=
  {n | Odd n \land \neg RepresentableByPrimeAndTwoPowers n}

/-- Any representation ‘p + 2^k + 2^l‘ is at least ‘4‘. -/
theorem representable_lower_bound {n p k l : \N} (hp : Nat.Prime p)
    (h : n = p + 2 ^ k + 2 ^ l) : 4 \le n := by
  subst h
  have hp2 : 2 \le p := hp.two_le
  have hk : 1 \le 2 ^ k := by exact Nat.one_le_two_pow
  have hl : 1 \le 2 ^ l := by exact Nat.one_le_two_pow
  omega

theorem not_representable_one : \not RepresentableByPrimeAndTwoPowers 1 := by
  intro hrepr
  rcases hrepr with \langle p, k, l, hp, hrepr\rangle
  have h4 : 4 \le 1 := representable_lower_bound hp hrepr
  omega

theorem one_mem_exceptionalOddSet : 1 \in ExceptionalOddSet := by
  constructor
  \cdot decide
  \cdot exact not_representable_one

/-- Every odd natural number below ‘4‘ is exceptional for this representation shell. -/
theorem odd_lt_four_mem_exceptionalOddSet {n : \N} (hodd : Odd n) (hn_lt : n < 4) :
    n \in ExceptionalOddSet := by
  constructor
  \cdot exact hodd
  \cdot intro hrepr
    rcases hrepr with \langle p, k, l, hp, hEq\rangle
    have h4 : 4 \le n := representable_lower_bound hp hEq
    omega

end PrimeTwoPowers
end AraLibrary
```

B Source

The formalization file is located at ara$_{l}ibrary/formal/AraLibrary/NumberTheory/PrimeTwoPowers.lean.$